

Program : Diploma in Renewable Energy	
Course Code: 4293	Course Title : Renewable Energy Engineering II
Semester : 4	Credits: 4
Course Category : Program Core	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- To identify different Wind energy conversion systems.
- To understand the production of energy from Bioresources and Fuel Cell.
- To understand the concepts of Tidal and Ocean Energy.
- To understand the concepts of Small Hydro and Geothermal energy sources.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Different renewable sources		Basic Electrical Engineering and Renewable Technology.	2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Identify different Wind energy conversion systems(WECS).	17	Applying
CO2	Outline the concepts of energy from Bioresources and Fuel cells.	14	Understanding

CO3	Summarize the production of energy from oceans and Tides.	14	Understanding
CO4	Illustrate the concepts of Small Hydro and Geothermal Power plants.	13	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3				3		
CO2	2				3		
CO3	2				3		
CO4	2				3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identify different Wind energy conversion systems(WECS)		
M1.01	Illustrate the basic components of WECS.	4	Understanding
M1.02	Identify the power available from the wind.	5	Applying
M1.03	Classify various types of wind turbines.	4	Understanding
M1.04	Illustrate different schemes of WECS.	4	Understanding

Contents:

Wind Energy: Introduction- characteristics of Wind -Factors affecting distribution of wind energy on the surface of earth-Wind Energy Conversion Systems (WECS) -Basic principle-Advantages and disadvantages -Site selection considerations- -Basic components- block diagram - Classification.

Power and Energy from Wind -Basic terms and definitions(Aerodynamics, aerofoil,Lift, drag) - Power and Energy from Wind (Derivation and problems)-Power versus wind speed characteristics-Capacity Factor-Wind energy conversion efficiency-Coefficient of performance-Wind energy storage-Application of wind power-Wind energy program in India.

Classification of wind turbines- Introduction- classification- Based on type of rotor- Based on orientation of rotor -Schematic diagram of Horizontal and vertical axis turbines- comparison between horizontal axis and vertical axis turbines- Wind turbine operation-Wind speed - Basic terms (Actual speed, rated speed , cut in speed , cut out speed)

Schemes for electric power generation-Types of generators used - Principle of induction generators -Torque speed characteristics - Schemes for electric power generation-Block diagrams- Constant speed constant frequency - Variable speed constant frequency- Variable speed variable frequency.

CO2	Outline the concepts of energy from Bio resources and Fuel cells		
M2.01	Summarize the concept of energy production from bio resources.	3	Understanding
M2.02	Explain various biomass conversion technologies for energy production.	4	Understanding
M2.03	Classify various Biogas plants.	4	Understanding
M2.04	Illustrate the working of fuel cells.	3	Understanding
	Series Test – I	1	

Contents :

Bio Energy- Energy from Biomass and biogas -basic concepts - Photosynthesis process- Usable forms of biomass and their composition and fuel properties- Biomass resources- Factors affecting generation of Biogas.

Biomass- Biomass conversion Technologies-Physical methods,direct combustion , Thermochemical method,Biochemical method,Wet process,Dry process -Biomass Gasification-Classification of Biomass Gasifiers- Fixed Bed Gasifiers (Downdraft , Updraft ,Cross draft Gasifiers) -Fluidized Bed Gasifier-(Schematic Diagram and brief explanation Only)

Biogas-Biogas production from Biomass-Anaerobic digestion system (Block diagram)- Advantages of Anaerobic Digestion- Biogas plants-Classification -Continuous and Batch types- Dome and Drum types- Construction of typical biogas plants-Floating drum type (KVIC model) -Fixed Dome type (JANATA model).

Fuel Cell- Principle of operation-Classifications(listing only) -Hydrogen and Oxygen fuel cells(alkaline fuel cells, Proton Exchange Membrane fuel cells) - Advantages Limitations -Applications- Differences between battery and fuel cells.

CO3	Summarize the production of energy from oceans and Tides		
M3.01	Summarize the concept of ocean energy.	2	Understanding
M3.02	Classify various Tidal power plants.	5	Understanding
M3.03	Explain the principle of wave energy conversion systems.	4	Understanding
M3.04	Illustrate various types of Ocean Thermal Energy Conversion plants.	3	Understanding

Contents :

Ocean Energy-Introduction-Ocean energy resources(tides,waves and ocean -thermal)

Tidal Energy- Introduction-Basic terms (Barrage,flood tide ,ebb tide, tidal ranges) Basic principle-Tidal technologies (Tidal stream technologies,Tidal range technologies)- Classification of Tidal power plants-Tidal turbine farms- Tidal barrages (Single basin arrangement,Double basin arrangement)-Advantages and limitations

Energy from Ocean Waves- Introduction-Basic terms (crest,trough,amplitude, wavelength)- Wave energy conversion devices-Floats-High level Reservoir Wave machine-Dolphin Type Wave power Machine-Hydraulic Accumulator Wave Machine-Oscillating water column (schematic diagram and working only)-Advantages and limitations.

Ocean Thermal Energy Conversion systems(OTEC) -Principle of operation-Types of OTEC plants - Open cycle- Closed cycle- Hybrid cycle (Block diagram and description only)- Site selection criteria-Advantages and limitations

CO4	Illustrate the concepts of Small Hydro and Geothermal power plants		
M4.01	Illustrate the layout of a Small-Hydro Power Station.	3	Understanding
M4.02	Classify various types of turbines in small hydro Power Plants.	3	Understanding
M4.03	Explain different geothermal resources.	4	Understanding
M4.04	Illustrate the schematic diagram of geothermal power plants.	3	Understanding
	Series Test – II	1	

Contents :

Small Hydro Resources-Introduction-Nature of small hydro development-Classifications (based on head - capacity- scheme)-Lay out of a Small-Hydro scheme- Advantages and Disadvantages-Applications-Criterion for selection of sites.

Water Turbines: Basic concepts-types of turbines-Francis-Kaplan-Pelton wheel-Chain turbine- Water wheels- Bulb turbine-Tube turbine.

Geothermal Energy-Introduction-Nature of geothermal fields-Types of geothermal sources-(Hydrothermal resources,Hot Dry Rock resources,Geo pressured resources,Magma resource)- Advantages and limitations-Applications -Geothermal exploration-Geothermal well drilling-Operational and Environmental problems-Geothermal energy in India.

Geothermal power plants- Classification -schematic diagram and working of Vapor dominated geothermal power plant-Liquid dominated geothermal power plant (Flash steam, Open system and Binary cycle system).

Text / Reference

T/R	Book Title/Author
T1	G.D Rai - Non conventional Energy Sources, Khanna Publishers
T2	Khan B H-Non conventional Energy Resources, Tata McGraw Hill, 2009
R1	Sawhney .G.S Non Conventional Energy Resources, PHI Learning, 2012

Online Resources

Sl.No	Website Link
1	https://openei.org/wiki/Wind_energy
2	https://www.nrel.gov/research/re-biomass.html

3	https://www.yourelectricalguide.com/2018/05/tidal-power-plant-working-principle.html
4	https://otecorporation.com/otec
5	https://www.irena.org/geothermal
6	https://www.youngscientistlab.com/sites/default/files/interactives/wind-energy/
7	Fuel cell Technology https://nptel.ac.in/courses/103/102/103102015/